

Level 1: Array Creation

1.

Create a NumPy array containing:

[10, 20, 30, 40, 50]

2.

Create an array of numbers from 1 to 20.

3.

Create an array of even numbers from 2 to 50.

4.

Create an array of 10 zeros.

5.

Create an array of 15 ones.

6.

Create an array:

[5, 5, 5, 5, 5, 5]

without typing 5 six times.

Level 2: Array Properties

Given:

```
arr = np.array([[1,2,3],  
               [4,5,6],  
               [7,8,9]])
```

7.

Print:

shape

size

ndim

dtype

8.

Find total number of elements.

9.

Find number of rows and columns.

Level 3: Indexing & Slicing

Given:

```
arr = np.array([10,20,30,40,50,60,70,80])
```

10.

Print:

30

11.

Print:

[20,30,40]

12.

Print last 3 elements.

13.

Reverse the array using slicing.

Level 4: Array Operations

Given:

```
a = np.array([1,2,3,4,5])
```

```
b = np.array([10,20,30,40,50])
```

14.

Add both arrays.

15.

Subtract both arrays.

16.

Multiply both arrays.

17.

Divide b by a.

18.

Square every element of a.

19.

Find cube of every element of a.

Level 5: Mathematical Functions

Given:

```
arr = np.array([5,10,15,20,25])
```

20.

Find:

sum

mean

max

min

21.

Find median.

22.

Find standard deviation.

23.

Find index of maximum element.

24.

Find index of minimum element.

Level 6: 2D Arrays

25.

Create:

```
[[1 2 3]
```

```
[4 5 6]
```

```
[7 8 9]]
```

26.

Print second row.

27.

Print third column.

28.

Print:

5

29.

Print:

```
[[1 2]
```

```
[4 5]]
```

30.

Find sum of each row.

Level 7: Reshape

31.

Create array:

```
[1,2,3,4,5,6,7,8,9,10,11,12]
```

Convert it into:

3×4

32.

Convert same array into:

2×6

33.

Flatten:

```
[[1,2,3],
```

```
[4,5,6]]
```

into:

```
[1,2,3,4,5,6]
```

Level 8: Boolean Masking

Given:

```
arr = np.array([12,45,7,89,34,22,100])
```

34.

Print all elements greater than 30.

35.

Print all even numbers.

36.

Count how many numbers are greater than 50.

37.

Replace all numbers less than 20 with 0.

Level 9: Mini Challenges

38.

Find second largest number in a NumPy array.

39.

Remove duplicate values from:

[1,2,2,3,4,4,5,6,6]

40.

Check whether an array contains a particular number entered by user.

41.

Count frequency of each element.

42.

Find all prime numbers between 1 and 100 using NumPy.